Industrial Internship Report on

"Smart City Traffic Pattern Analysis and Forecasting"

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Domain: Data Science and Machine Learning

upskill Campus | The IoT Academy | UniConverge Technologies Pvt Ltd (UCT)

Executive Summary

This report presents the Industrial Internship completed under upskill Campus and The IoT Academy, in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT), in the domain of Data Science and Machine Learning.

The internship began with a structured learning phase covering the foundations of data science, probability and statistics, and machine learning, each validated through assessments, before moving into an applied project.

The assigned project, based on the “Smart City Traffic Patterns” dataset from Kaggle, involves analyzing and forecasting traffic volumes at four city junctions to support smart infrastructure planning, accounting for variation across normal days, holidays, and special occasions.

Overall, this internship provided valuable exposure to real-world data science problems, strengthened core technical foundations, and gave hands-on experience translating theoretical concepts into a practical forecasting solution.

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Preface

This report summarizes my progress during the Data Science and Machine Learning internship offered through upskill Campus (USC) and The IoT Academy, in association with UniConverge Technologies Pvt Ltd (UCT).

A relevant industrial internship is an important step in career development — it bridges the gap between classroom learning and real-world application, and gives early exposure to how data-driven problem solving works in an industry setting.

My assigned project is on Smart City Traffic Pattern Analysis and Forecasting, based on a public dataset from Kaggle, where the goal is to help a city government understand traffic behavior at four junctions and plan infrastructure proactively, accounting for holidays and special occasions.

USC and UCT structured this opportunity as a phased program: the initial weeks were dedicated to building strong theoretical foundations in data science, statistics, and machine learning, each reinforced with assessments, before progressing into the applied project phase.

Weekly progress across the foundation-building phase is summarized below:

- Week 2: Studied "Introducing Data Science" (Cielen, Meysman & Ali), covering the data science lifecycle, data preparation, visualization, and ML fundamentals; scored 100% on the associated quiz.
- Week 3: Studied "An Introduction to Probability and Statistics" (Shewhart & Wilks), covering probability, descriptive statistics, random variables, distributions, sampling, hypothesis testing, and statistical inference.
- Week 4: Studied machine learning fundamentals through a video lecture and "Introduction to Machine Learning" (Smola & Vishwanathan), covering the ML workflow, supervised and unsupervised learning, preprocessing, feature engineering, and model evaluation; scored 90% on the associated quiz.

This structured foundation directly supports the project phase, where these statistical and machine learning concepts are applied to real traffic data.

I am grateful to the mentors and team at USC, The IoT Academy, and UCT for their guidance and for structuring this learning journey so effectively. My message to juniors and peers taking up this internship: invest genuine effort in the foundation weeks — the statistics and ML concepts covered early on make the project phase far more approachable.

Introduction

About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT), established in 2013, works in the Digital Transformation space, delivering industrial solutions with a focus on sustainability and return on investment.

UCT builds its products and solutions on cutting-edge technologies including the Internet of Things (IoT), Cyber Security, Cloud Computing (AWS, Azure), Machine Learning, communication technologies (4G/5G/LoRaWAN), Java Full Stack, Python, and front-end development.

Its offerings include the UCT Insight IoT platform for rapid deployment of IoT applications, the Factory Watch smart-factory platform for production and asset monitoring, LoRaWAN-based solutions for agritech, smart cities, and utility metering, and predictive maintenance solutions that combine embedded systems, industrial IoT, and machine learning.

About upskill Campus (USC)

upskill Campus, together with The IoT Academy and in association with UniConverge Technologies, facilitated the smooth execution of this internship program.

USC is a career development platform that delivers personalized, executive-style coaching in a way that is affordable, scalable, and measurable.

The IoT Academy

The IoT Academy is the EdTech division of UCT, running long-form executive certification programs in collaboration with EICT Academy and IITK, IITR, and IITG across multiple domains.

Objectives of this Internship Program

- To get practical experience of working in the industry.
- To solve real-world problems using data science and machine learning.
- To improve job prospects through applied, hands-on work.
- To build a stronger understanding of the field and its applications.
- To support personal growth, including communication and problem-solving skills.

Reference

[1] Cielen, D., Meysman, A. D. B., & Ali, M. – Introducing Data Science.

[2] Shewhart, W. A., & Wilks, S. S. – An Introduction to Probability and Statistics.

[3] Smola, A., & Vishwanathan, S. V. N. – Introduction to Machine Learning.

[4] Kaggle Dataset – Smart City Traffic Patterns: https://www.kaggle.com/utathya/smart-city-traffic-patterns

Glossary

Term	Acronym
Data Science	DS
Machine Learning	ML
Exploratory Data Analysis	EDA
Root Mean Squared Error	RMSE
Mean Absolute Error	MAE
AutoRegressive Integrated Moving Average	ARIMA
Long Short-Term Memory (network)	LSTM
Key Performance Indicator	KPI

Problem Statement

The city government is working to transform the city into a smart, digitally-enabled city that improves the efficiency of citizen services. Traffic congestion is one of the key challenges standing in the way of this vision.

As a data scientist on this initiative, the task is to help the government better manage city traffic and provide input into future infrastructure planning. Specifically, the goal is to understand and forecast traffic patterns at four major junctions in the city.

Traffic behavior on holidays and other special occasions differs meaningfully from that on normal working days, and this variation must be accounted for when building a forecasting model — so that the city is prepared for traffic peaks rather than reacting to them.

Existing and Proposed Solution

Traffic management in most cities today still relies heavily on manual counts, fixed-timing signals, and reactive interventions once congestion is already visible. These approaches offer little advance warning and do not adapt to recurring patterns such as holidays or seasonal events.

The proposed solution is a data-driven forecasting model built on historical junction-level traffic data from the Kaggle “Smart City Traffic Patterns” dataset. The model will learn recurring time-based patterns (hour of day, day of week, month) as well as the effect of holidays and special occasions, to forecast expected traffic volume at each of the four junctions.

The value addition is a shift from reactive to proactive traffic management — giving the city government advance insight into expected peaks so that signal timing, resource deployment, and infrastructure investment can be planned ahead of time rather than in response to congestion.

Code Submission (GitHub link)

https://github.com/anshhh5/upskillcampus/blob/main/smartcitytrafficpatternanalysisandforecasting.py

Report Submission (GitHub link)

https://github.com/anshhh5/upskillcampus/blob/main/SmartCityTrafficPatternAnalysisandForecasting-Anshjeet-USC-UCT%20(1).pdf

Proposed Design / Model

The solution follows a standard applied data science pipeline, moving from raw data to actionable forecasts:

Figure 1: Proposed data-to-forecast pipeline

High Level Diagram

Raw traffic data (per junction, timestamped vehicle counts) is collected from the Kaggle dataset, cleaned and preprocessed, enriched with calendar-based features (hour, weekday/weekend, holiday flag), used to train forecasting models, and finally used to generate junction-wise traffic forecasts and planning insights, as shown in Figure 1.

Low Level Diagram

[To be added once feature engineering and model architecture are finalized during the project implementation phase.]

Interfaces

Planned interfaces include the raw CSV data ingestion layer, a preprocessing/feature-engineering module, the model training and evaluation module, and an output layer producing forecast values and visualizations for each junction. Details of data flow and any APIs will be added as the implementation progresses.

Performance Test

Since this is a forecasting problem intended for real infrastructure planning — not just an academic exercise — evaluating model accuracy and reliability under realistic constraints is essential.

Key constraints identified include forecasting accuracy across normal versus holiday/special-occasion periods, and the ability of the model to generalize to unseen time periods rather than only fitting historical data.

Test Plan / Test Cases

A time-based train-test split is planned, holding out the most recent period of data (rather than a random split) to realistically simulate forecasting future, unseen traffic. Separate evaluation is planned for normal days versus holidays/special occasions.

Test Procedure

Candidate models (e.g., ARIMA/SARIMA as a time-series baseline, and tree-based models such as Random Forest or Gradient Boosting, alongside an LSTM-based approach) will be trained on the historical portion of the data and evaluated on the held-out period for each junction.

Performance Outcome

[To be added after model training and evaluation are completed in the next phase of the project — planned metrics are RMSE and MAE per junction, compared across candidate models.]

My Learnings

The foundation-building weeks of this internship significantly strengthened my technical base for data science work. Studying “Introducing Data Science” gave me a clear picture of the end-to-end data science lifecycle — from data preparation and visualization through to applying machine learning — and scoring 100% on the related quiz confirmed a solid grasp of these fundamentals.

The following week's deep dive into probability and statistics, though the most challenging — particularly hypothesis testing and statistical inference — built the mathematical intuition that underlies most data science and machine learning techniques.

Studying machine learning fundamentals in Week 4, through both video lectures and reference texts, clarified the ML workflow, the distinction between supervised and unsupervised learning, and how preprocessing, feature engineering, and evaluation metrics fit together; a 90% quiz score reflected good conceptual understanding, while also highlighting areas — particularly the mathematical intuition behind certain algorithms and metric selection — that need continued practice.

Together, these foundation weeks gave me the confidence and technical grounding to approach the Smart City Traffic Patterns project, and I now see clearly how probability, statistics, and ML concepts translate into a real forecasting problem with practical, real-world stakes.

Future Work Scope

- Complete exploratory data analysis on the Smart City Traffic Patterns dataset and finalize feature engineering (holiday calendars, special-event flags).
- Implement and compare multiple forecasting approaches (ARIMA/SARIMA, tree-based models, LSTM) and select the best-performing model per junction.
- Build a simple dashboard to visualize forecasted vs. actual traffic for city planners.
- Extend the model with additional real-world signals such as weather data or live sensor feeds, if available.
- Explore scaling the approach beyond four junctions to a city-wide traffic forecasting solution.